

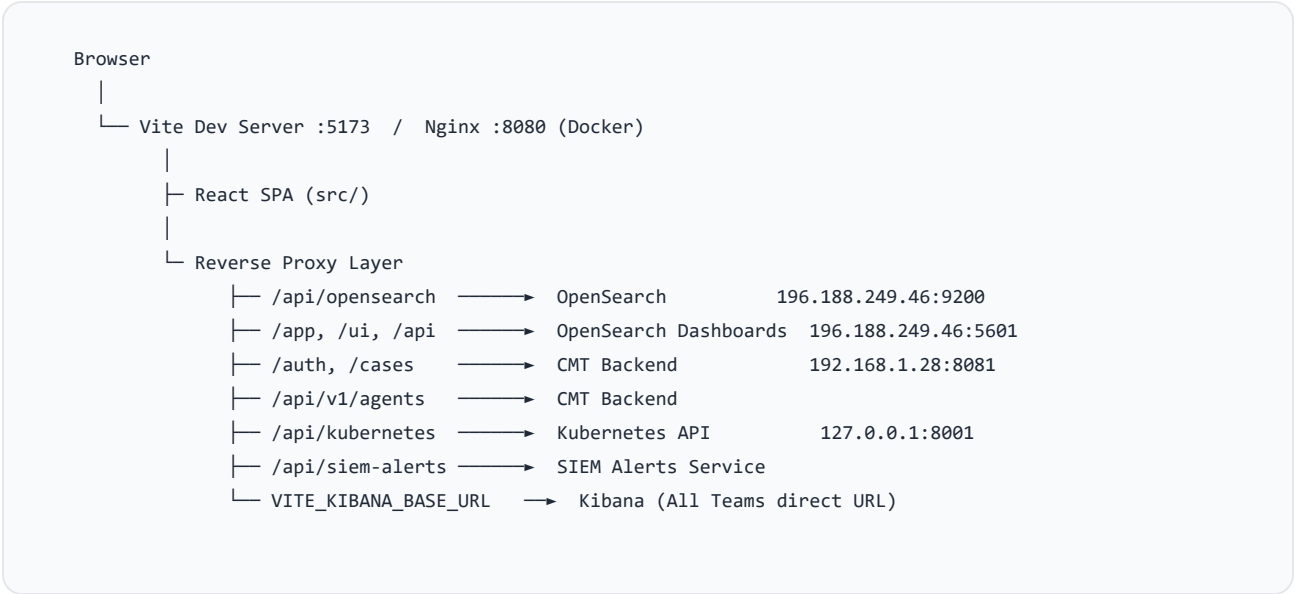
TECHNICAL ARCHITECTURE DOCUMENT

BETA Security Dashboard

Unified SOC Platform — System Architecture Overview

STACK	BACKEND	AUTH	VERSION
React 18 + Vite + OpenSearch	196.188.249.46:9200	Keycloak OIDC + PKCE	OpenSearch 2.13 / Kibana- ready

1 System Overview



● **Frontend**

- React 18 + Vite
- Tailwind CSS + Radix UI
- Recharts for visualizations
- Same-origin iframe embedding
- Lucide icons

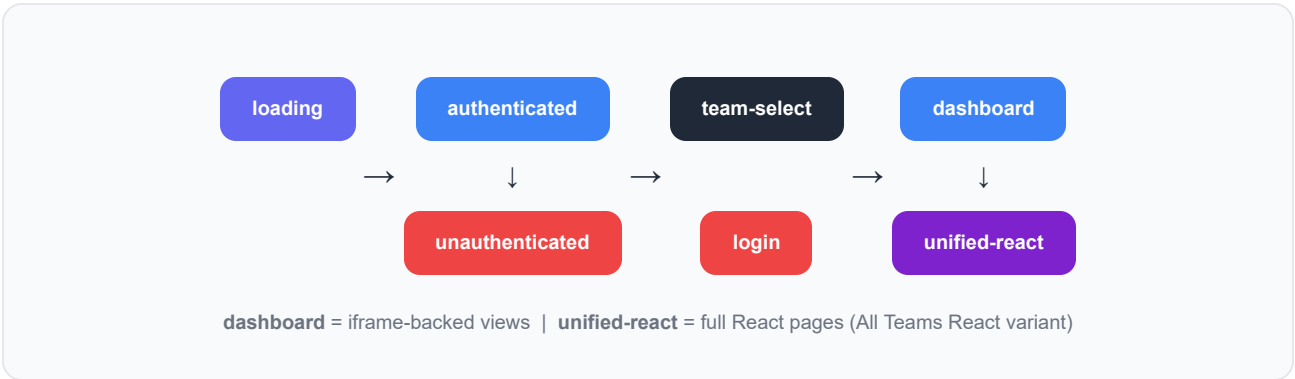
● **Data Layer**

- OpenSearch 2.13.0 (:9200)
- OpenSearch Dashboards (:5601)
- Kafka + Zookeeper (:9092)
- Vector data pipeline
- Kibana (optional, unified)

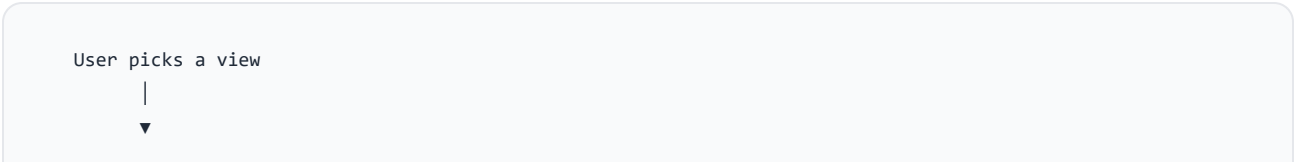
● **Security**

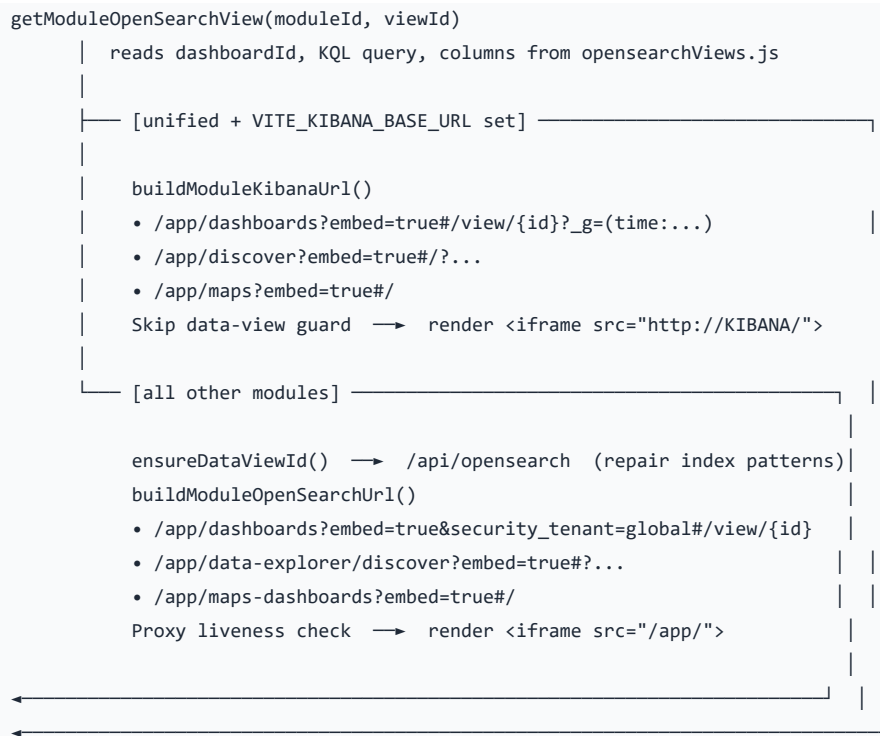
- Keycloak OIDC + PKCE
- Bearer token on all CMT calls
- Scope-based access control
- sessionStorage token store
- Multi-tenant via Kubernetes NS

2 App State Machine (App.jsx)



3 iframe Rendering Pipeline





4 Source Tree

```

src/ ── App.jsx ← Main router + app state machine ── main.jsx ← React DOM entry point | ── auth/ | ──
keycloak.js ← OIDC + PKCE flow, token in sessionStorage | ── accessControl.js ← Scope/permission checks,
API token management | ── config/ | ── moduleConfig.js ← Menu structure for all 5 modules | ──
opensearchViews.js ← Per-view KQL queries, dashboard IDs, URL builders | ── api/ | ── opensearch.js ←
Direct OpenSearch queries (/api/opensearch proxy) | ── opensearchDashboards.js ← Data view (index pattern)
repair & lookup | ── cmt.js ← Case management & ticketing (Keycloak-authed) | ── edr.js ← EDR data from
OpenSearch indices | ── siemAlerts.js ← Live SIEM alert stream (SSE) + case creation | ──
responseActions.js ← Manual analyst actions | ── securityServices.js ← Playbooks, approvals, threat
hunting, audit | ── kubernetes.js ← Kubernetes API for tenant management | ── otelApi.js ← OpenTelemetry
logs/traces/metrics | ── notifications.js ← Alert notification delivery | ── components/ | ── Auth/ | ──
Login.jsx ← Login form UI | ── Onboarding/ | ── TeamSelector.jsx ← Team/tenant picker shown after login | ──
Layout/ | ── Navbar.jsx ← Top bar (time range, search, user menu) | ── Sidebar.jsx ← Left nav (module menu
items) | ── AlertCenter.jsx ← Toast notification center | ── ManualResponseDialog.jsx ← Incident response
actions dialog | ── SocOperationsDialog.jsx ← SOC workflow panel | ── Dashboard/ | ── IndexOverview.jsx ←
OpenSearch index browser & stats | ── SecurityOverview.jsx ← High-level security metrics | ── Pages/ | ──
SiemAlerts.jsx ← SIEM triage (React page, not iframe) | ── CmtDashboard.jsx ← Case management (React page,
not iframe) | ── UnifiedReactDashboard.jsx ← React-based unified view | ── ObservabilityDashboard.jsx ←
OTel logs/traces/metrics | ── SocAutomationPage.jsx ← Playbook/automation workflows | ── Tenants.jsx ←
Multi-tenant administration | ── [EdrDashboard, Endpoints, Events, Malware].jsx | ── ui/ ← Radix UI +
Tailwind design system | ── alert, badge, button, card, dialog, input, select, skeleton, table ...

```

5 Module System

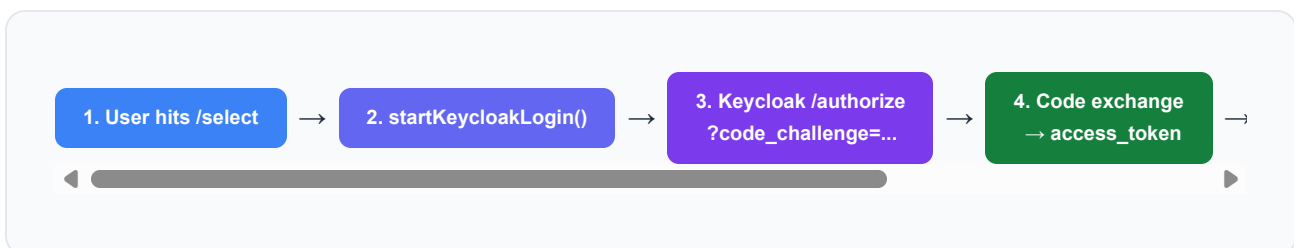
MODULE	ROUTE	IFRAME SOURCE	INDEX PATTERNS	NOTES
siem	/siem	OpenSearch Dashboards	wazuh-alerts-4.x-*, wazuh-*	Wazuh-indexed SIEM events
ids	/ids	OpenSearch Dashboards	edr*, logs-*, wazuh-*	Suricata / Zeek IDS data
edr	/edr	OpenSearch Dashboards	edr*, edr-detections-*, tenant-*-edr*	EDR detections & response actions
unified	/unifild	Kibana (if VITE_KIBANA_BASE_URL set) or OpenSearch	All of the above combined	All Teams — recently migrated to Kibana

MODULE	ROUTE	IFRAME SOURCE	INDEX PATTERNS	NOTES
observability	/observability	OpenSearch Dashboards	opensearch_dashboards_sample_data_otel_*	OTel logs / traces / metrics

6 Vite Proxy Routes (vite.config.js)

<code>/api/opensearch/*</code>	→ OpenSearch 196.188.249.46:9200 always on
<code>/api/kubernetes/*</code>	→ Kubernetes API 127.0.0.1:8001 K8s required
<code>/api/v1/agents/*</code>	→ CMT Backend 192.168.1.28:8081 optional
<code>/api/v1/*</code>	→ Kubernetes API (catch-all after /agents match)
<code>/api/siem-alerts/*</code>	→ SIEM Alerts Service optional
<code>/auth, /cases, /alerts, /users, /webhooks, /ingest</code>	→ CMT Backend root routes
<code>/api/playbooks/*</code>	→ Playbook Service optional
<code>/api/response/*</code>	→ Response Action Service optional
<code>/api/approvals/*</code>	→ Approval Queue Service optional
<code>/api/threat-hunting/*</code>	→ Threat Hunting Service optional
<code>/api/audit-service/*</code>	→ Audit Service optional
<code>/app, /ui, /bootstrap.js, /startup.js, /translations, /internal, /goto, /api</code>	→ OpenSearch Dashboards 196.188.249.46:5601 (iframe routes)

7 Authentication Flow (Keycloak PKCE)



● keycloak.js — Key Functions

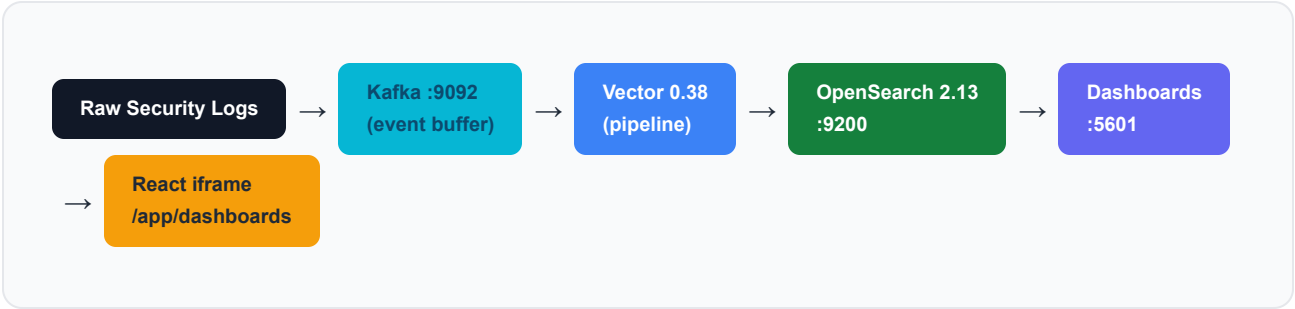
- `getKeycloakConfig()` — Read from `window._env_` or `import.meta.env`
- `startKeycloakLogin()` — OIDC auth code + PKCE redirect
- `initializeKeycloakSession()` — Handle callback, exchange code
- `getKeycloakAccessToken()` — Return current token
- `logoutFromKeycloak()` — Logout + redirect
- `clearKeycloakSession()` — Clear `sessionStorage`

● accessControl.js — Key Functions

- `getConfiguredApiToken()` — Read from `localStorage`
- `setConfiguredApiToken()` — Store to `localStorage`
- `getConfiguredScopes()` — Extract from token + storage
- `hasRequiredScopes()` — Permission check
- `missingRequiredScopes()` — Return what's missing
- `scopeMatches()` — Wildcard scope matching (admin:*)

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Data Pipeline (Docker Stack)



DOCKER SERVICE	IMAGE	PORT	ROLE
opensearch	opensearchproject/opensearch:2.13.0	9200	Search & analytics engine
dashboards	opensearchproject/dashboards:2.13.0	5601	Visualization UI
dashboards-bootstrap	python:3.11	—	Init script — import saved objects
zookeeper	confluentinc/cp-zookeeper:7.5.0	2181	Kafka coordinator
kafka	confluentinc/cp-kafka:7.5.0	9092, 29092	Event streaming
kafka-ui	provectuslabs/kafka-ui	9090	Kafka admin UI
vector	timberio/vector:0.38.0	9598 (metrics)	Log routing & transformation
security-dashboard	custom (nginx)	8080	Built React app + Nginx proxy

9 API Client Layer (src/api/)

● opensearch.js

- Direct OpenSearch cluster queries
- `GET /api/opensearch/_cat/indices`
- `GET /api/opensearch/_search`
- `GET /api/opensearch/{index}/_doc/{id}`

● opensearchDashboards.js

- Data view (index pattern) management
- Auto-repair missing saved objects
- Called before iframe renders
- Supports multi-title fallback

● cmt.js — Case Management

- Keycloak Bearer token authed
- SSE live update support
- Routes: `/auth`, `/cases`, `/alerts`, `/report-templates`, `/users`, `/webhooks`
- Fallback: `/api/v1/agents`

● siemAlerts.js

- Real-time alert stream (SSE)
- Case creation from triage
- Alert acknowledgement
- Enabled via `VITE_SIEM_ALERTS_ENABLED`

● edr.js

- EDR data from OpenSearch
- Index: `edr*`, `edr-detections-*`, `tenant-*-edr*`
- Detections, response actions, playbooks
- Audit events, forensic data

● securityServices.js

- Playbook orchestration
- Approval queue management
- Threat hunting campaigns
- Audit service log export

● kubernetes.js

- Namespace management for multi-tenancy
- `/api/kubernetes/api/v1/namespaces`
- mTLS for direct cluster access
- Kind cluster auto-detection

● otelApi.js

- OpenTelemetry logs, traces, metrics
- Index: `opensearch_dashboards_sample_data_otel_*`
- Service map data
- Distributed trace correlation

10 Environment Variable Reference (.env)

VARIABLE	DEFAULT	PURPOSE
<code>VITE_OPENSEARCH_PROXY_TARGET</code>	196.188.249.46:9200	OpenSearch cluster endpoint
<code>VITE_DASHBOARDS_PROXY_TARGET</code>	196.188.249.46:5601	OpenSearch Dashboards (iframes)
<code>VITE_KIBANA_BASE_URL</code>	(unset)	Kibana host for Unified/All Teams iframes
<code>VITE_CMT_PROXY_TARGET</code>	192.168.1.28:8081	Case management backend
<code>VITE_CMT_AUTO_CONNECT</code>	true	Auto-connect CMT on startup
<code>VITE_CMT_USE_KEYCLOAK_TOKEN</code>	true	Pass Keycloak token to CMT
<code>VITE_SIEM_ALERTS_ENABLED</code>	false	Enable live SIEM alert stream

VARIABLE	DEFAULT	PURPOSE
VITE_SECURITY_SERVICES_ENABLED	false	Enable playbook/response services
VITE_KUBERNETES_PROXY_TARGET	127.0.0.1:8001	Kubernetes API proxy
VITE_DEV_SERVER_HOST	localhost	Vite dev server bind address
VITE_DEV_SERVER_PORT	5173	Vite dev server port (strict)
VITE_KEYCLOAK_URL	http://localhost:8080	Keycloak server URL
VITE_KEYCLOAK_REALM	master	Keycloak realm name
VITE_KEYCLOAK_CLIENT_ID	vite-app	Keycloak client ID